



Data Warehouses, Data Lakes, and Lakehouses

(v1.00)

Week 2: September 15

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These slides are available at <https://lintool.github.io/cs431-2026f/>

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Key Questions

What are the main differences between operational and analytical infrastructure?

What are data warehouses?

What problems did they evolve to solve?

What are data lakes and lakehouses?

What problems did they evolve to solve?

What are the components of modern data platforms?

How do operational and analytical data models differ?

What goes on in ETL/ELT?

How do different physical representations of data affect storage, compute, and other tradeoffs within data platforms?

This Week

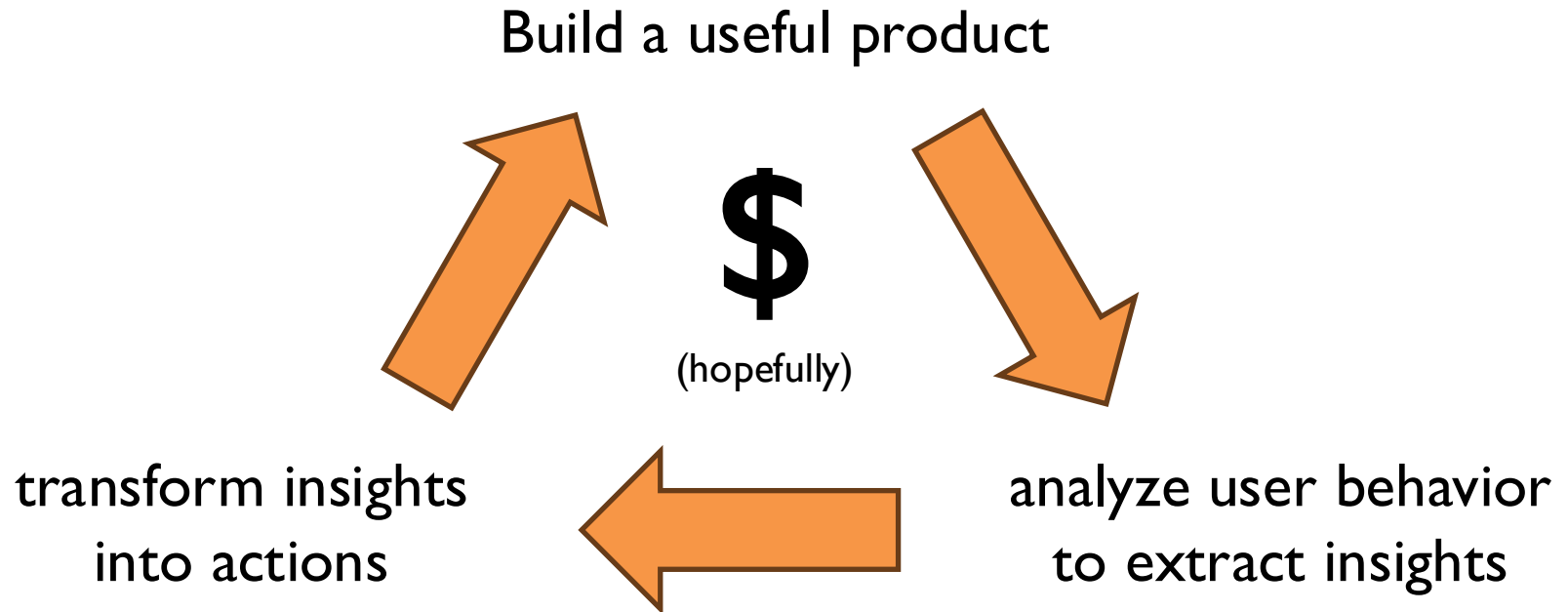
Now: Evolution of Data Platforms
Data Warehouses, Data Lakes, and Lakehouses

Next: Three Deep Dives
Data Modeling, ELT, Physical Representations

Context...

The Data Flywheel

(a virtuous cycle)



Google. Facebook. Twitter. Amazon. Uber.

Context...

What's this course about?

The *infrastructure* that supports the data flywheel.

data platforms + data engineering

Context...

What problems do data platforms solve?

Ingesting, storing, manipulating, maintaining, serving...
the data that supports the data flywheel.

Evolution of Data Platforms

Data Warehouses, Data Lakes, and Lakehouses

In the beginning...

users



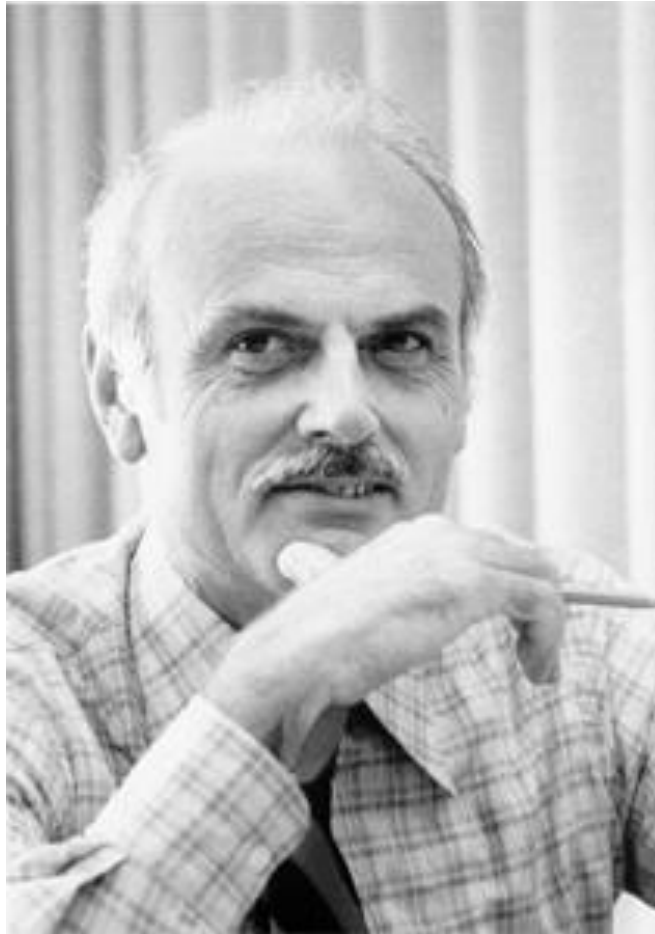
Monolithic
Application

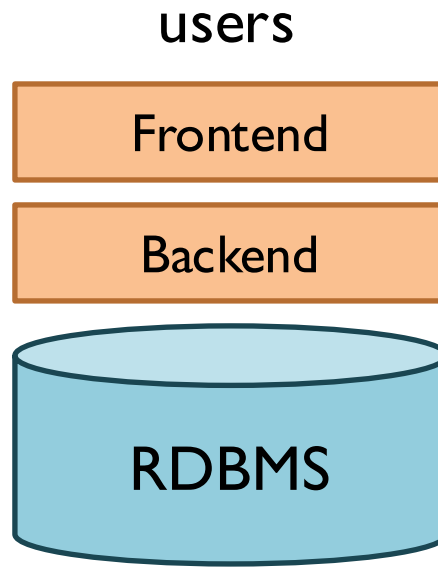
The diagram illustrates a monolithic application architecture. It features a central orange rectangular box with a dark brown border, containing the text "Monolithic Application". Above this box, the word "users" is written in black text. The entire diagram is set against a white background.

users

Frontend

Backend

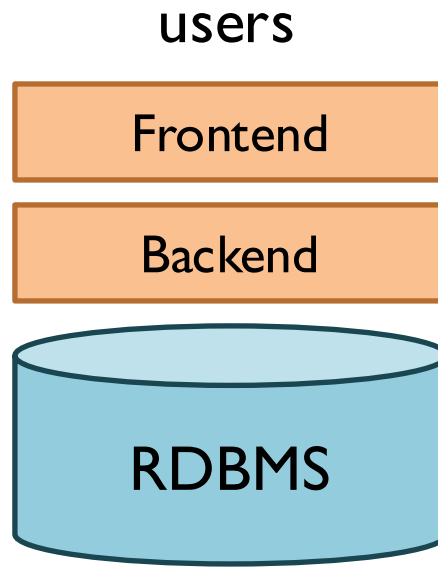




RDBMS = Relational Database Management System

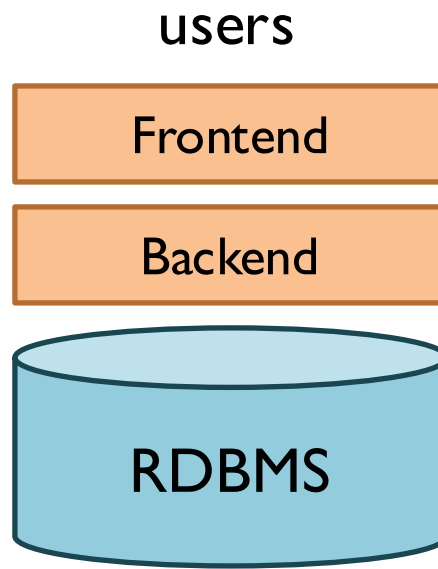
Imposes a relational view of data: tables, rows, columns

Provides a set of relational operators to manipulate data: SQL



Why is this a good idea?

- Offload physical data design
- Standardize query processing
- Ensure data integrity, manage concurrency
- Handle backup and recovery



Okay, but
why relational?

RDBMS = Relational Database Management System

Imposes a relational view of data: tables, rows, columns

Provides a set of relational operators to manipulate data: SQL

But we're constrained by the relational model?

(1) The relational model is surprisingly general.

(2) More later...

Remember from last time?

Business Intelligence: Case Study

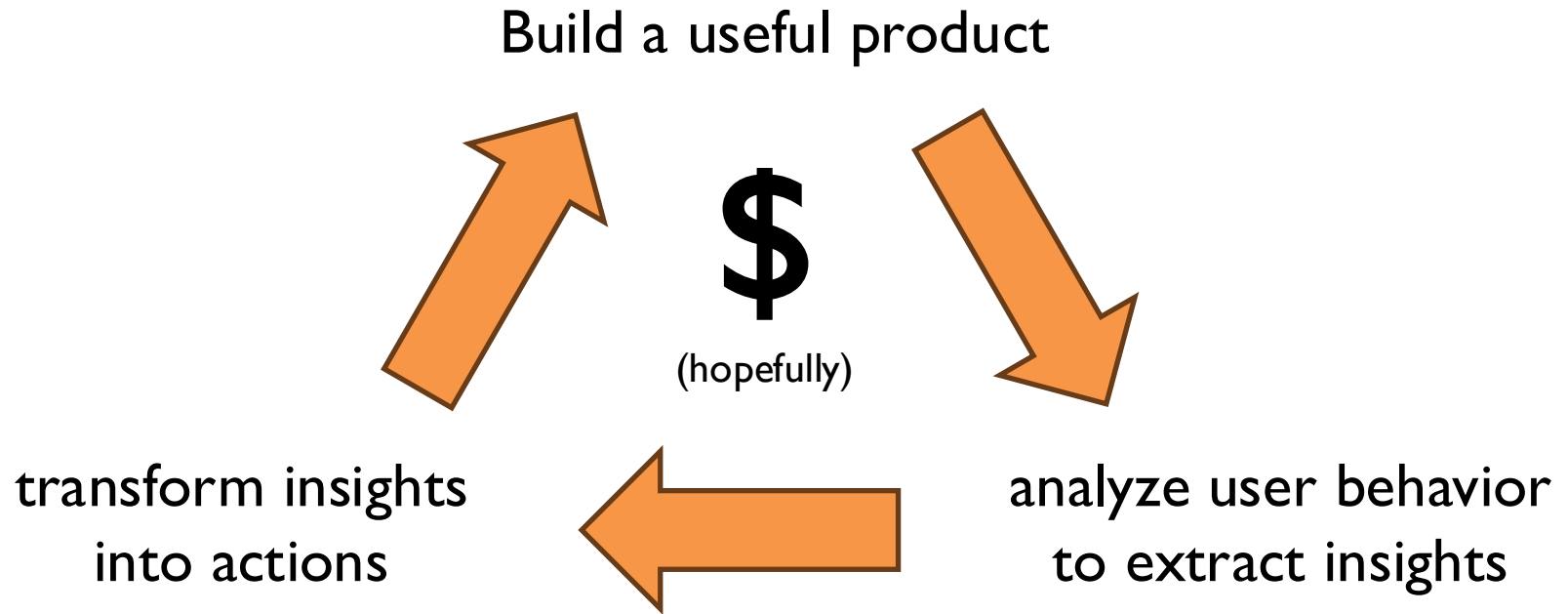
In the 1990s, Wal-Mart found that customers tended to buy diapers and beer together. So they put them next to each other and increased sales of both.*

* BTW, this is completely apocryphal. (But it makes a nice story.)

Remember from last time?

The Data Flywheel

(a virtuous cycle)



Google. Facebook. Twitter. Amazon. Uber.

users

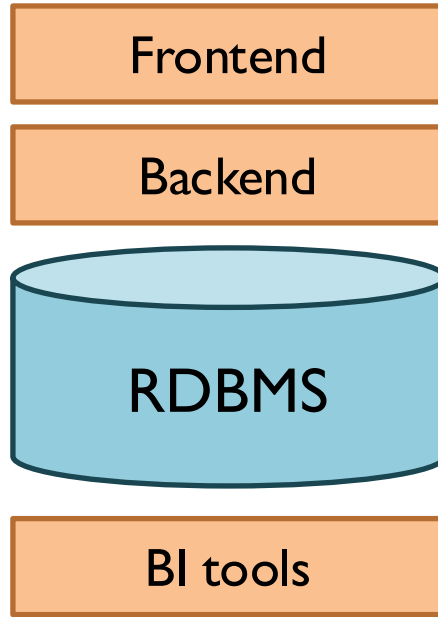
Frontend

Backend

RDBMS

BI tools

analysts



users

Frontend

Backend

RDBMS

BI tools

analysts



Why is my application so slow?

Why does my analysis take so long?



RDBMS Workloads

OLTP (online transaction processing)

Typical applications: e-commerce, banking, airline reservations

Customer-facing: real-time, low latency, highly concurrent

Tasks: relatively small set of transactional queries; CRUD

Data access pattern: random reads, updates, writes (small amounts of data)

OLAP (online analytical processing)

Typical applications: business intelligence, data mining

Back-end processing: batch workloads, less concurrency

Tasks: complex analytical queries, often ad hoc

Data access pattern: table scans, large amounts of data per query

OLTP and OLAP Together?

Downsides of co-existing OLTP and OLAP workloads

Conflicting queries and data access patterns

Poor memory management

Variable latency



users and analysts

Solution? Separate the two!



Build a data warehouse!

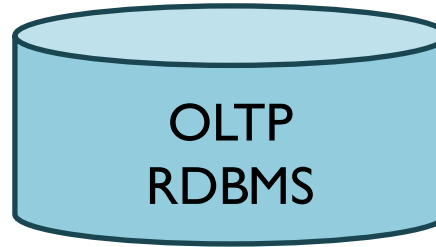


users

Frontend

Backend

OLTP database for
user-facing transactions



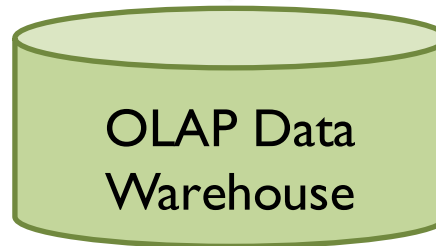
ETL

(Extract, Transform, and Load)

*Where and when does
this actually happen?*

Deep Dive later...

OLAP database for
data warehousing

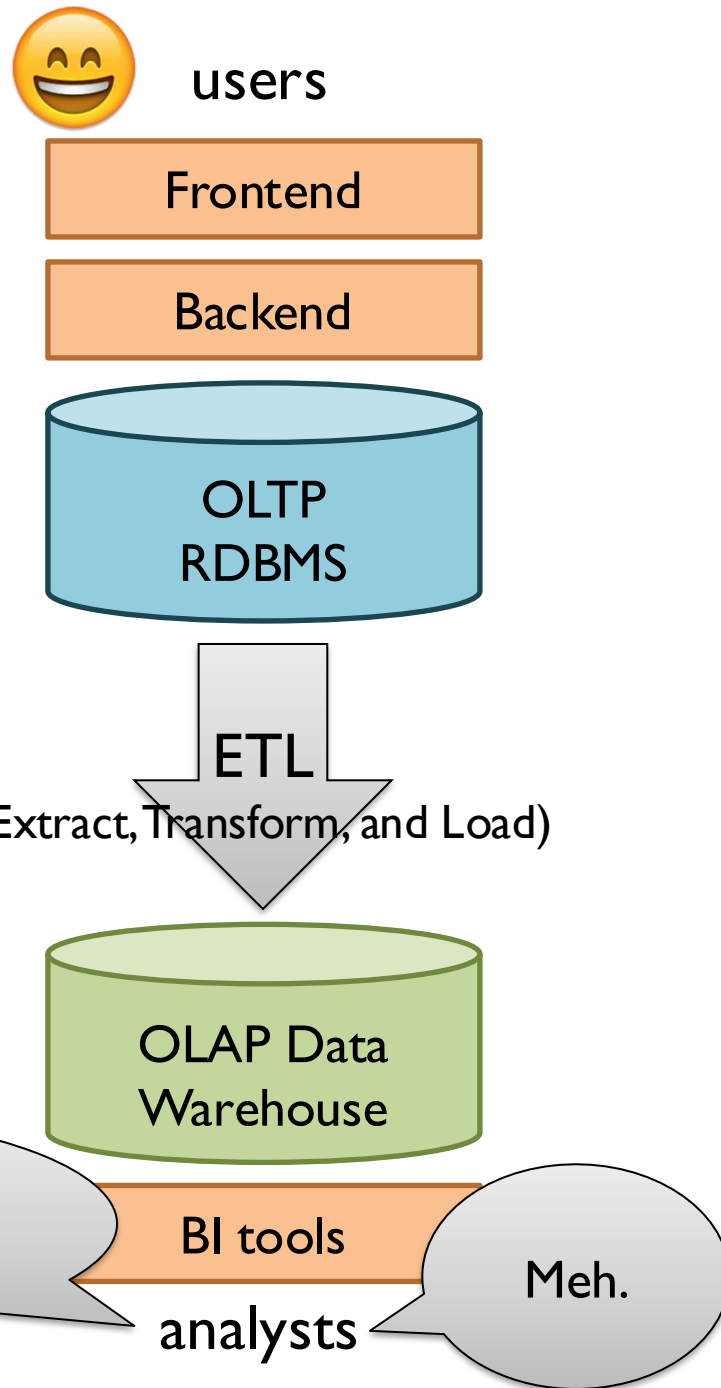


BI tools

analysts



Aside...

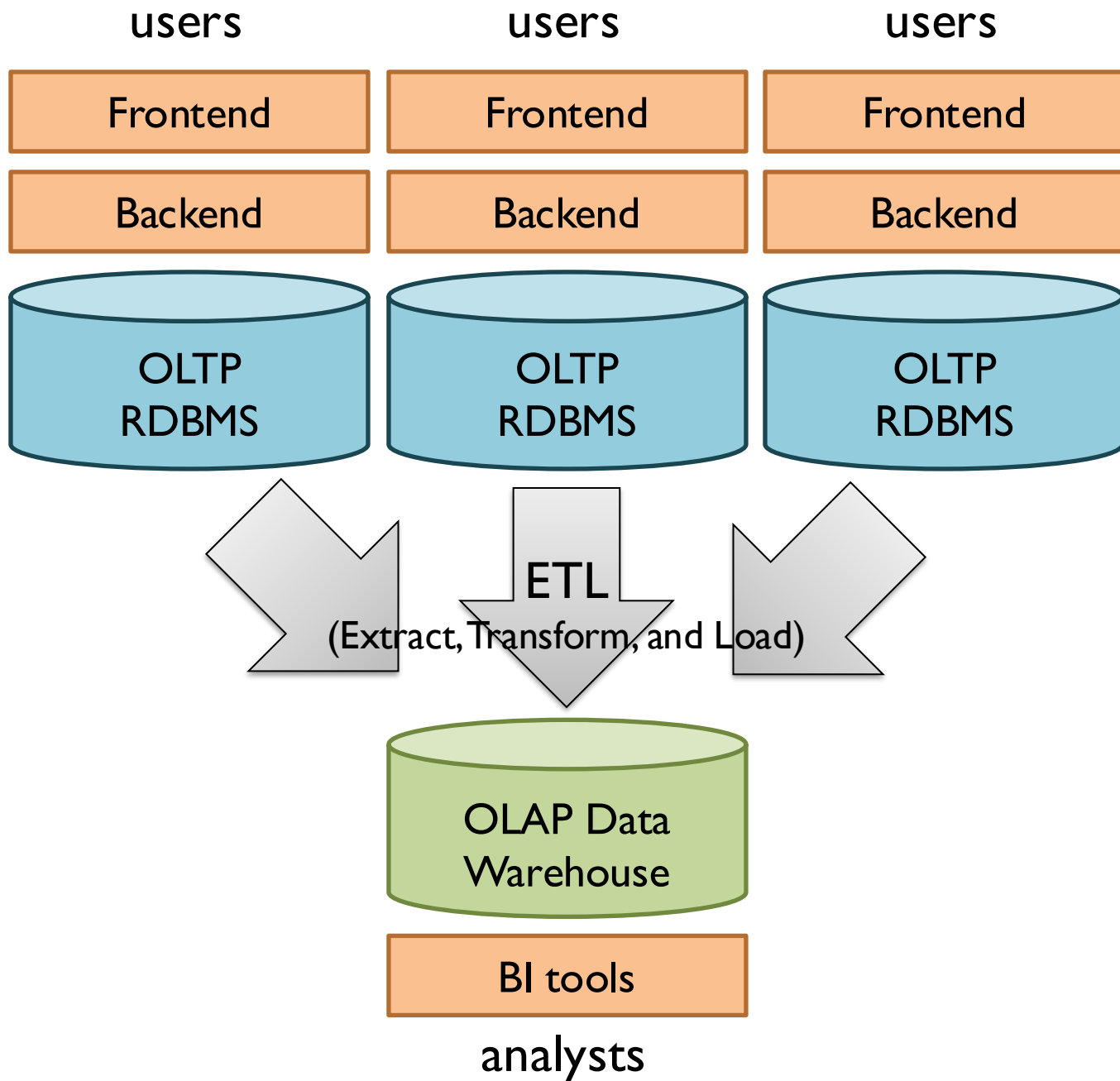


Aside...

Why is this a difficult problem?

Characteristics of the data: Volume, Velocity, Variety* + Veracity

* Coined by Gartner analyst Doug Laney in 2001.



facebook®

Jeff Hammerbacher, Information Platforms and the Rise of the Data Scientist.
In *Beautiful Data*, O'Reilly, 2009.

“On the first day of logging the Facebook clickstream, more than 400 gigabytes of data was collected. The load, index, and aggregation processes for this data set really taxed the Oracle data warehouse. Even after significant tuning, we were unable to aggregate a day of clickstream data in less than 24 hours.”

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users

Frontend

Backend

OLTP
RDBMS

Facebook context?

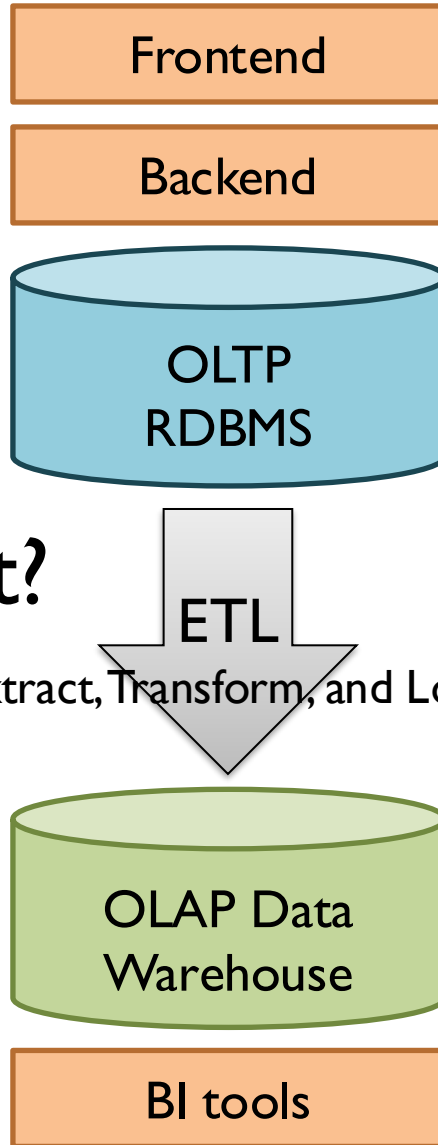
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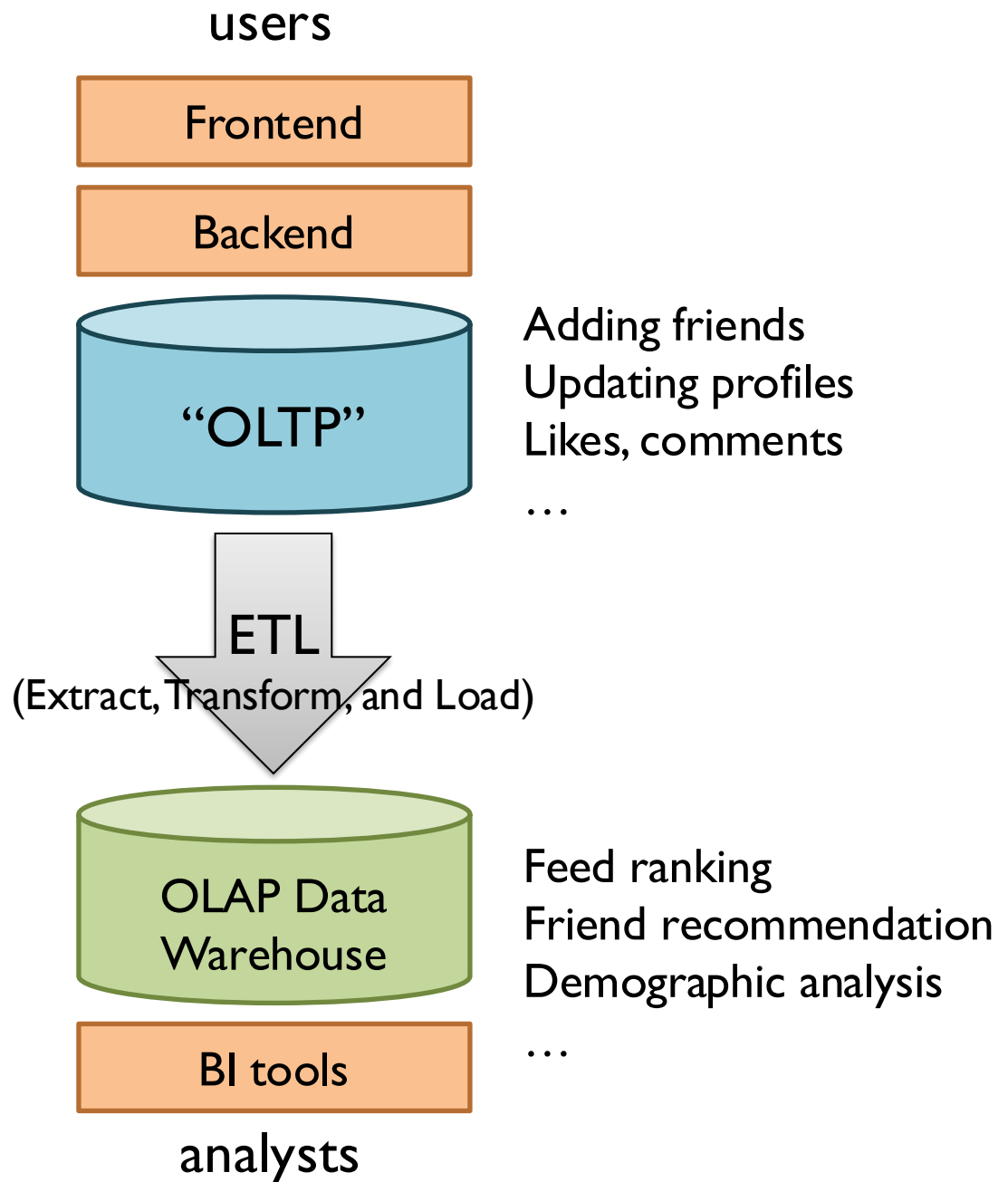
(Extract, Transform, and Load)

OLAP Data
Warehouse

BI tools

analysts





users

Frontend

Backend

“OLTP”

PHP/MySQL

ETL or ELT?

(Extract, Transform, and Load)

Hadoop



~~analysts~~

data scientists

users

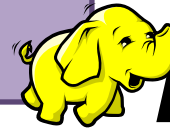
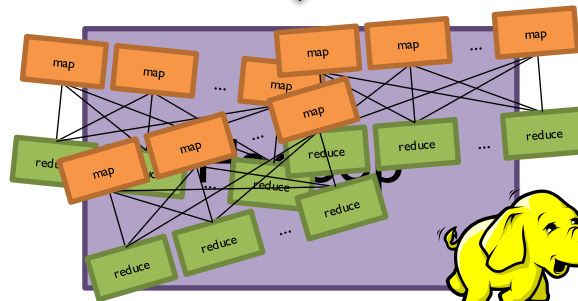
Frontend

Backend

“OLTP”

ELT

(Extract, Load, and Transform)



hadoop

data scientists

Data storage on HDFS
Processing with MapReduce

Why not just use an existing analytical database?

Scalability. Cost. Flexibility.

Jeff Hammerbacher, Information Platforms and the Rise of the Data Scientist.
In *Beautiful Data*, O'Reilly, 2009.

“On the first day of logging the Facebook clickstream, more than 400 gigabytes of data was collected. The load, index, and aggregation processes for this data set really taxed the Oracle data warehouse. Even after significant tuning, we were unable to aggregate a day of clickstream data in less than 24 hours.”

Databases are great...

If your data has structure (and you know what the structure is)

If your data is reasonably clean

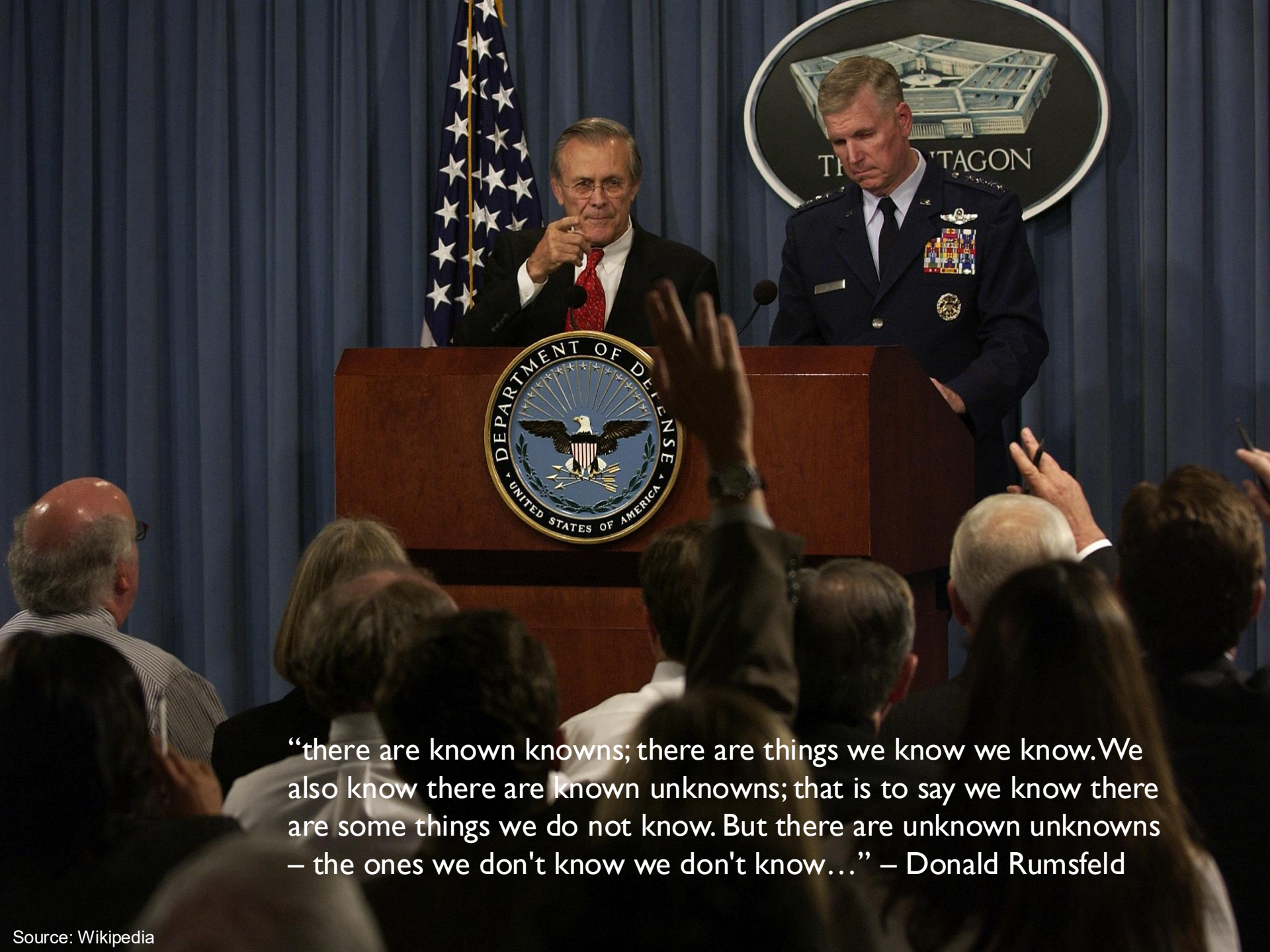
If you know what queries you're going to run ahead of time

Databases are not so great...

If your data has little structure (or you don't know the structure)

If your data is messy and noisy

If you don't know what you're looking for



“there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are unknown unknowns – the ones we don't know we don't know...” – Donald Rumsfeld

Databases are great...

If your data has structure (and you know what the structure is)

If your data is reasonably clean

If you know what queries you're going to run ahead of time

Known unknowns!

Databases are not so great...

If your data has little structure (or you don't know the structure)

If your data is messy and noisy

If you don't know what you're looking for

Unknown unknowns!

It'd be great...

If I could just ingest all my data
(relational, semi-structured, unstructured, graph, multimodal, etc.)
in a multitude of formats
(text, csv, json, etc.)
and figure out what to do with it later.



Data storage on HDFS

(later, Amazon S3, Google Cloud Storage, Azure Blob Storage, etc.)

If I could then process / manipulate / transform the data...
(I want to write *ad hoc* scripts **and** SQL queries)



Processing with MapReduce

(later, Spark, Presto, Flink, etc.)

tl;dr – I want *flexibility*.

What is a data lake?

A data lake is a central location that holds a large amount of data in its native, raw format... a data lake uses a flat architecture and object storage to store the data... By leveraging inexpensive object storage and open formats, data lakes enable many applications to take advantage of the data.

<https://www.databricks.com/discover/data-lakes>

A data lake is a centralized repository that ingests and stores large volumes of data in its original form. The data can then be processed and used as a basis for a variety of analytic needs. Due to its open, scalable architecture, a data lake can accommodate all types of data from any source, from structured... to semi-structured... to unstructured... The data files are typically stored in staged zones—raw, cleansed, and curated—so that different types of users may use the data in its various forms to meet their needs. Data lakes provide core data consistency across a variety of applications, powering big data analytics, machine learning, predictive analytics, and other forms of intelligent action.

<https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-data-lake>

What is a data lake?

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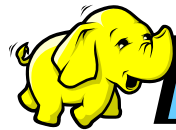
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<https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-data-lake>

Check?

It'd be great...

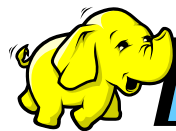
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hadoop Data storage on HDFS

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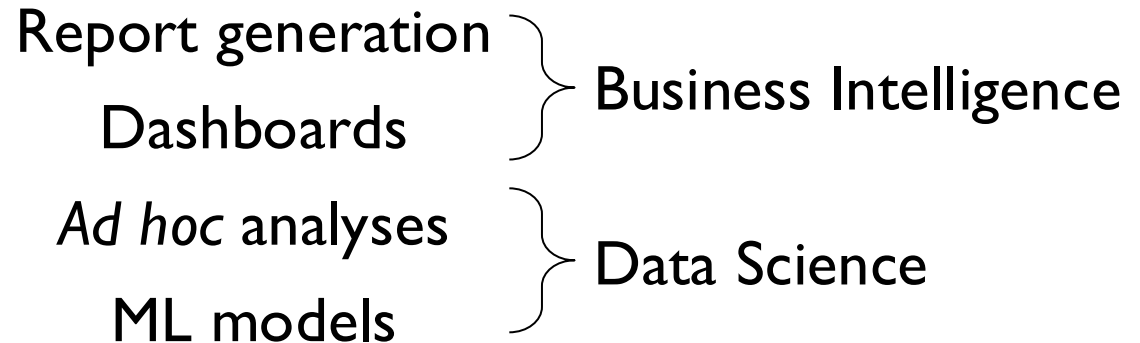
hadoop Processing with MapReduce

(later, Spark, Presto, Trino, Flink, etc.)

Reminder...

Transform Insights into Actions

What does that really mean?



known unknowns and unknown unknowns?

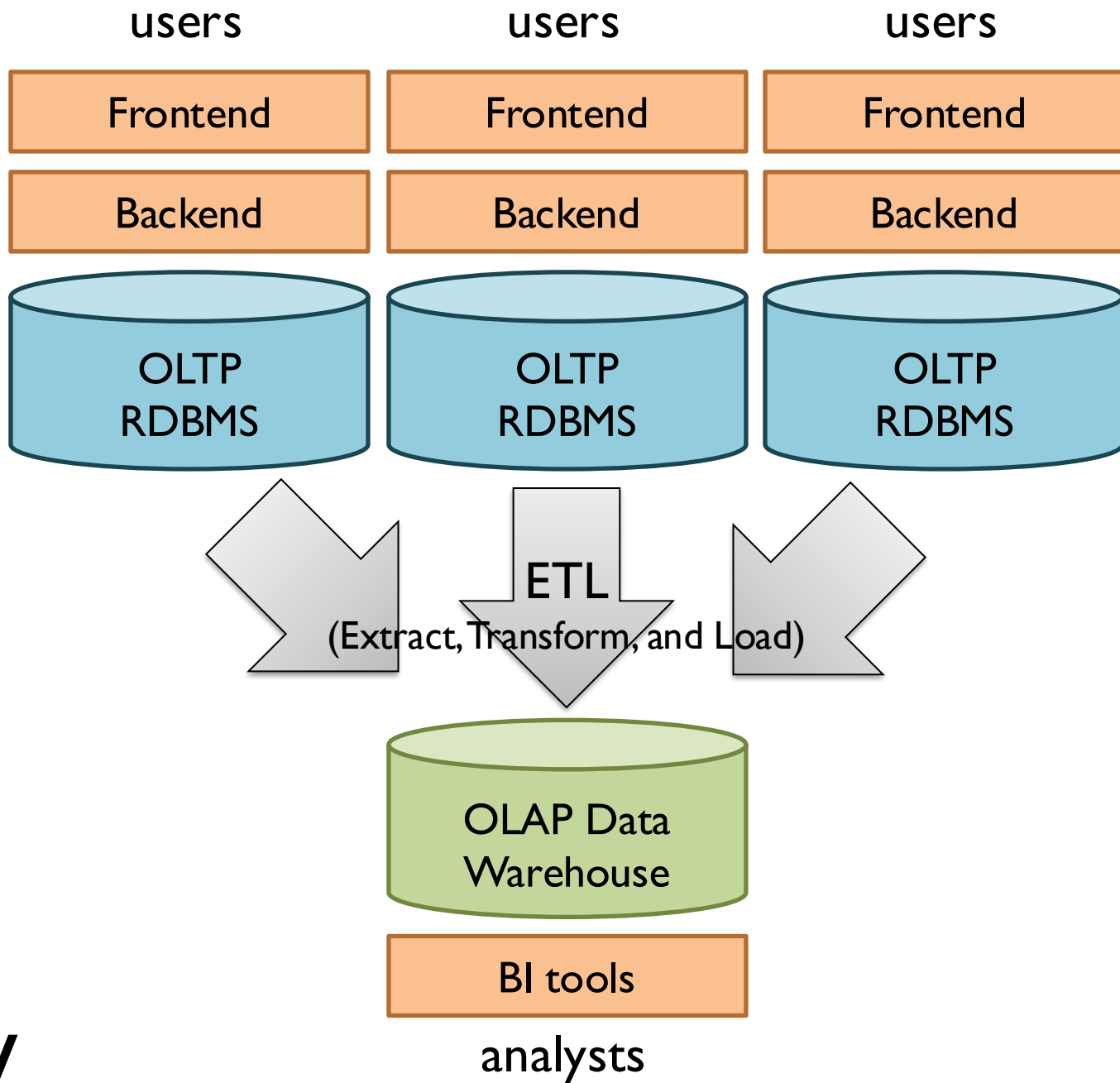
EDL vs. EDW

All types of data
vs. only relational data

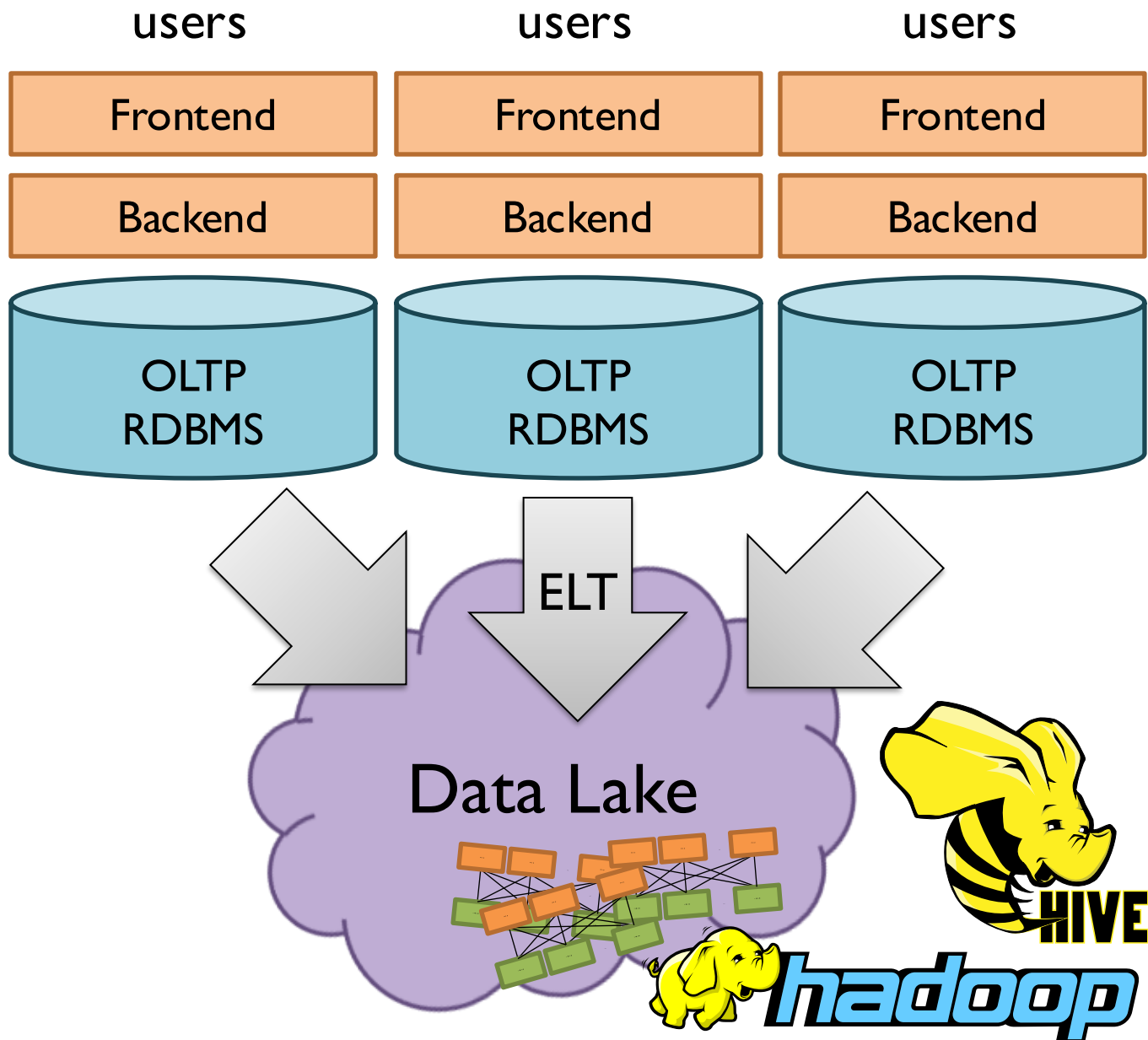
Open data formats
vs. proprietary formats

Cheap (object) storage
vs. expensive file systems

Flexible query processing
vs. only SQL



EDW



EDL (1st gen)

data scientists

Data Lakes to Data Swamps

tl;dr – I want *flexibility*.

Be careful what you wish for...

Why can't I find anything?

Where's the schema?

Why are there all these multiple near-duplicate copies?

Where did this come from?

Did this job actually finish?

Am I supposed to have access to this?

“jank”

The “Lakehouse” Vision

A unified metadata and governance layer over data lakes

Transaction support

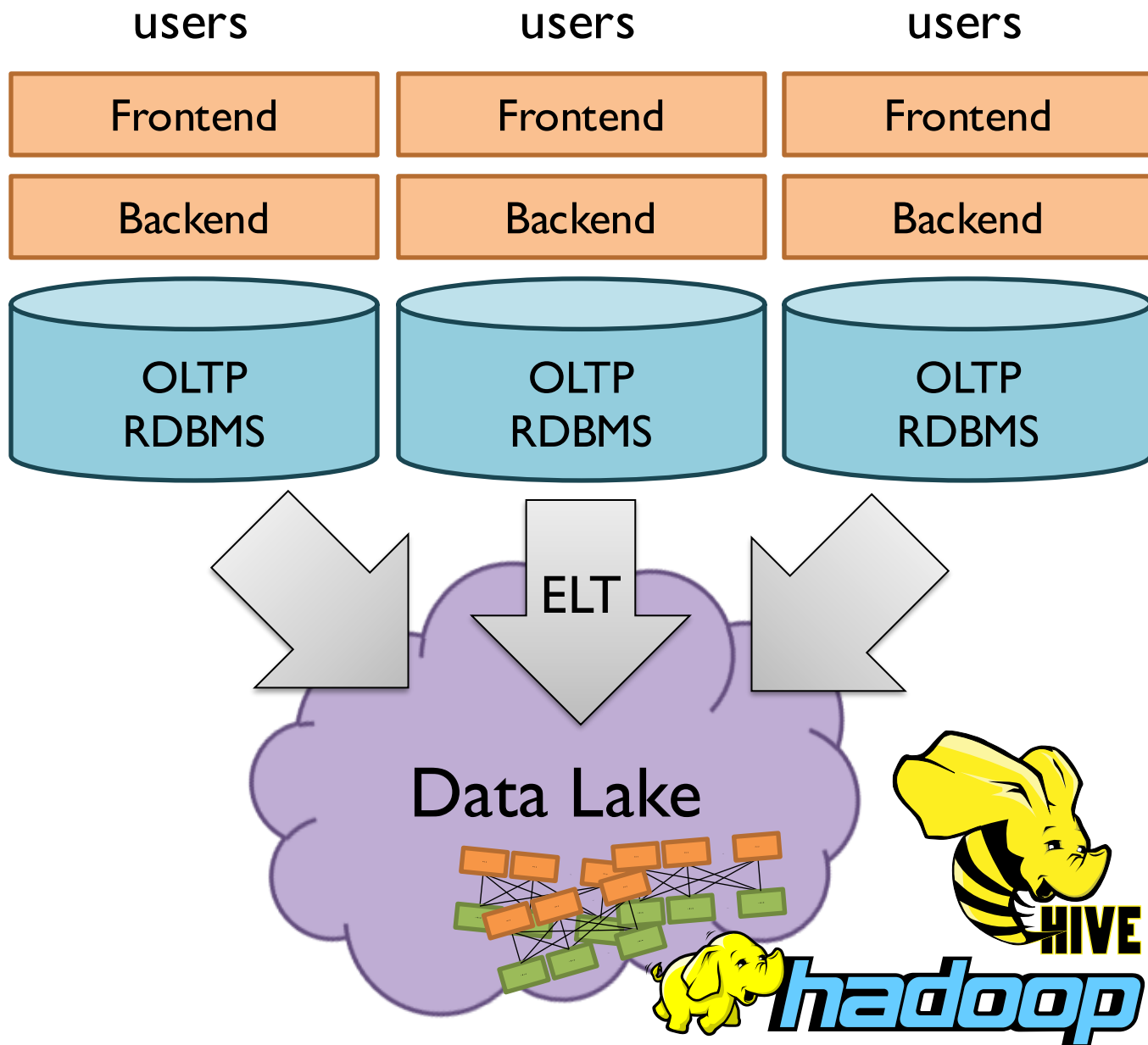
Schema enforcement and governance

Storage is decoupled from compute

Openness

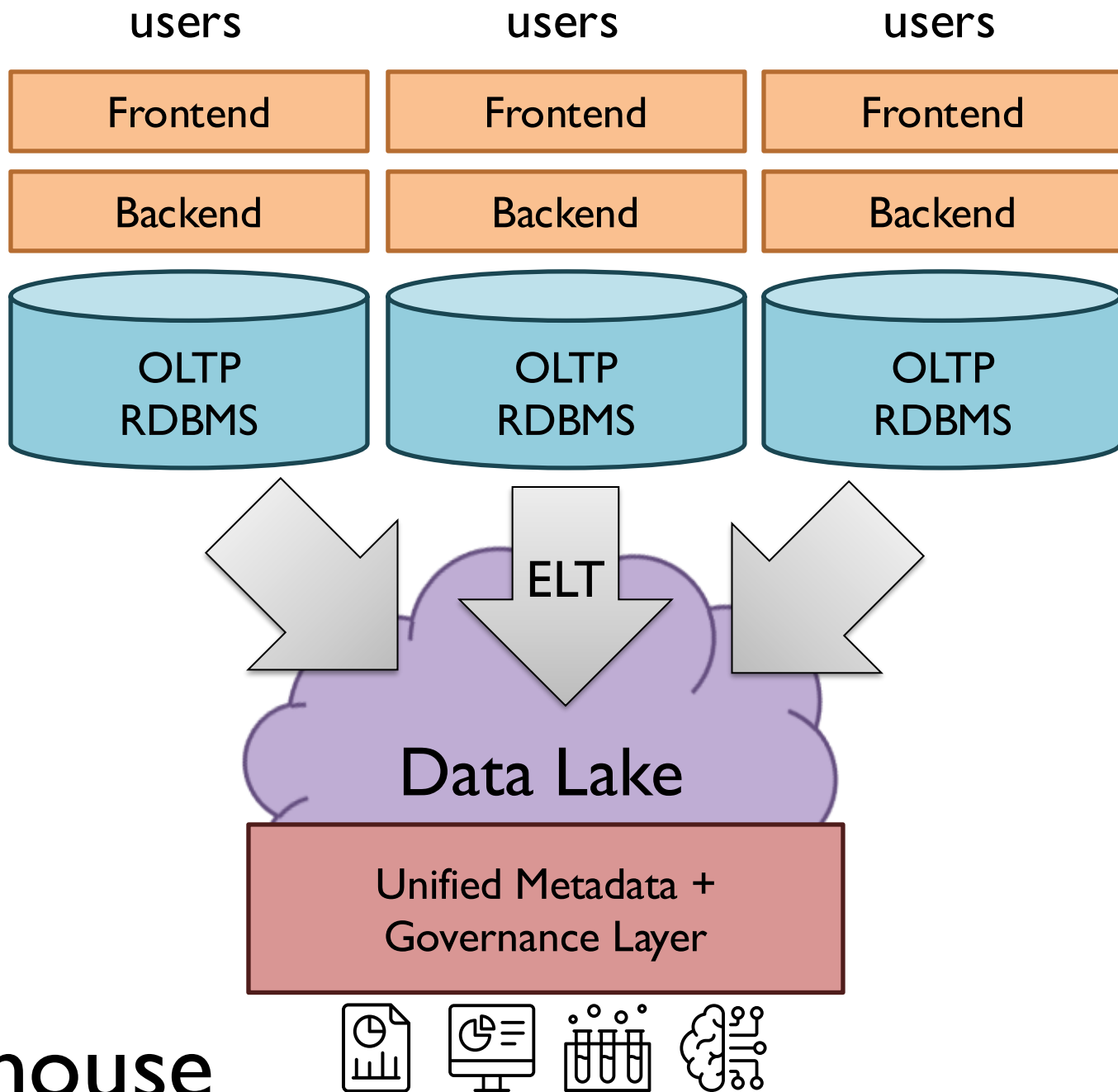
Support for diverse data types

Support for diverse workloads



EDL (1st gen)

data scientists



Lakehouse

EDL vs. EDW

Revenge of the DBs

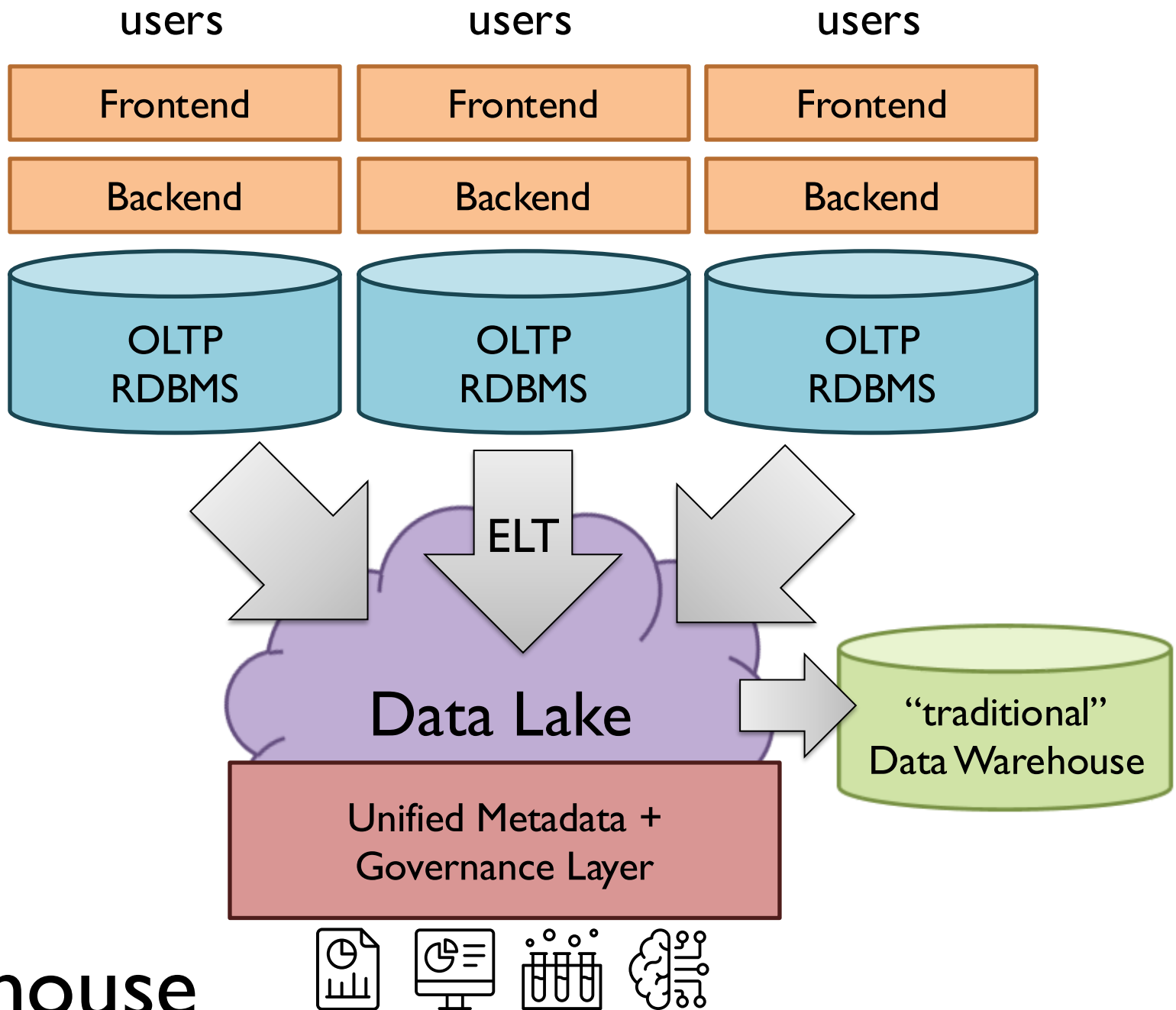
All types of data ✓
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Open data formats ✓
vs. proprietary formats

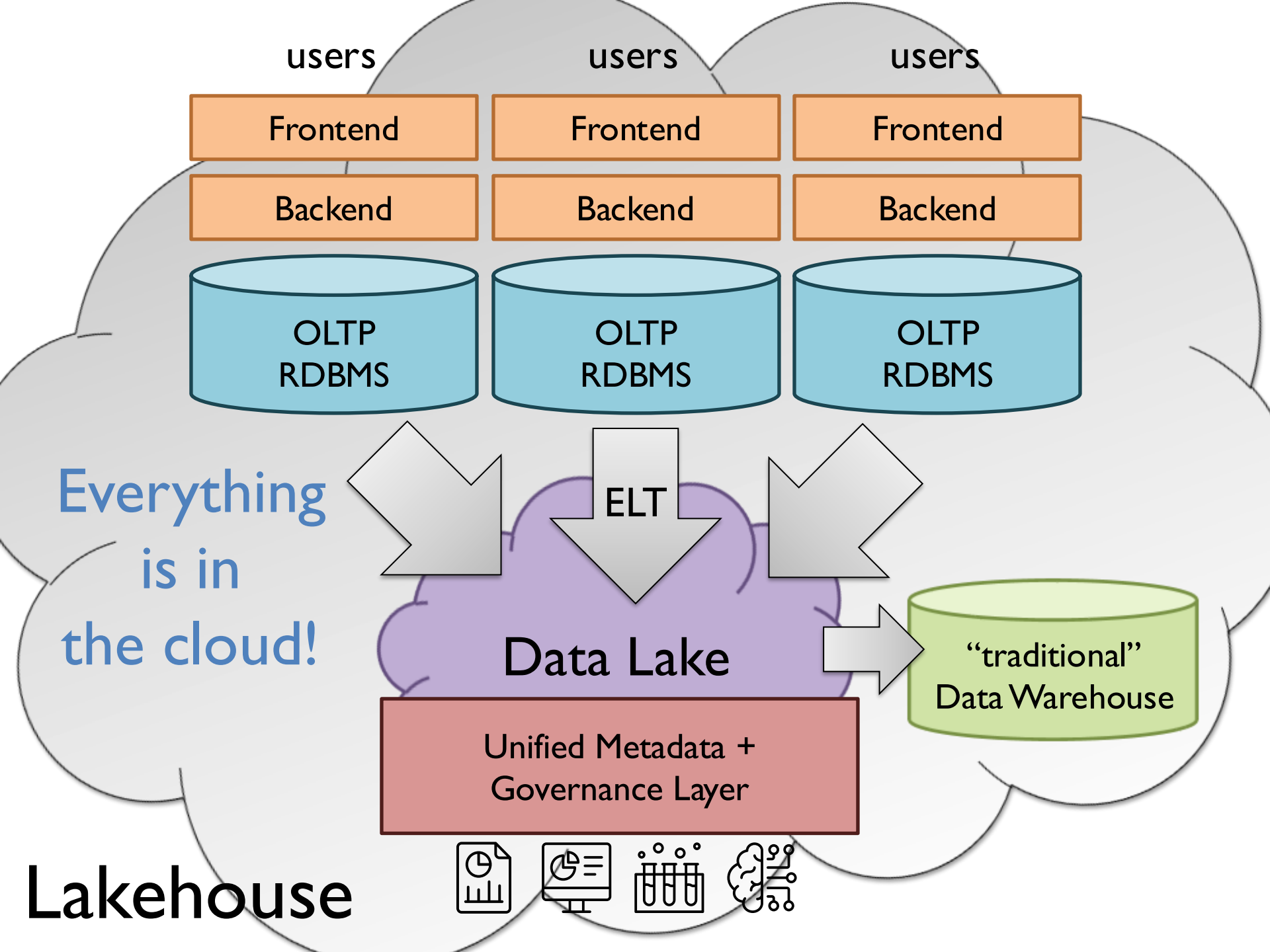
Cheap (object) storage ✓
vs. expensive file systems

Flexible query processing ✓
vs. only SQL

Sometimes, I really just want an RDBMS!



Lakehouse

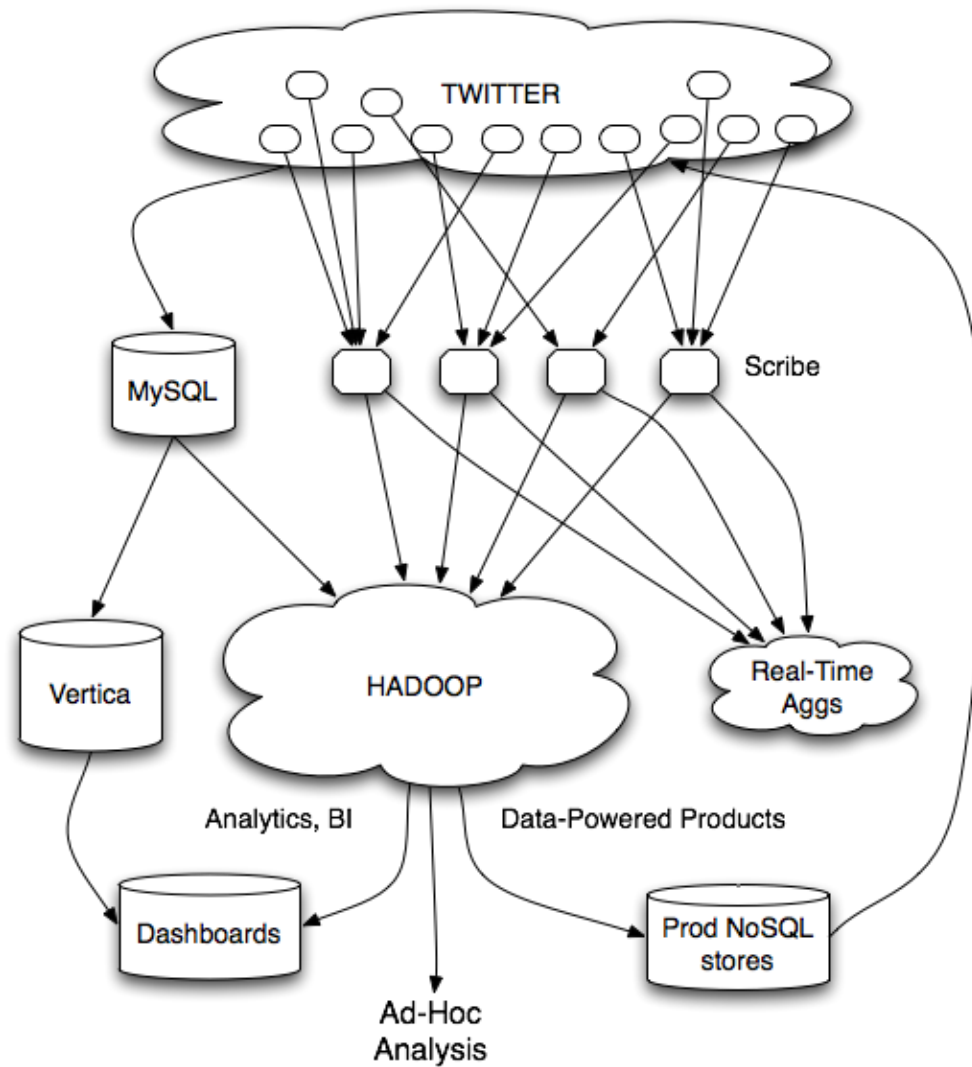


The Cloud

✓ Pro – you don't have to worry about it.

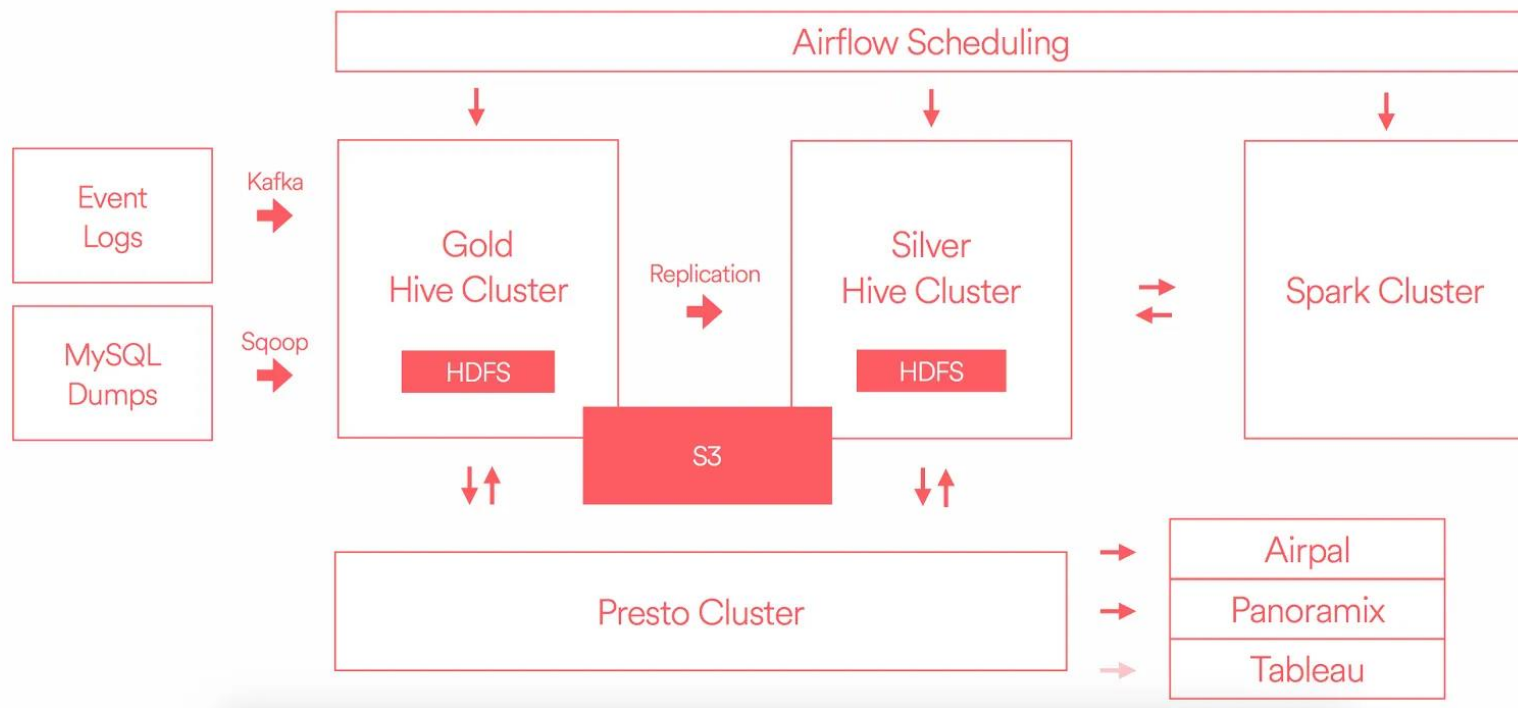
Con – you don't have to worry about it. ~~X~~

(More on this in the coming weeks...)



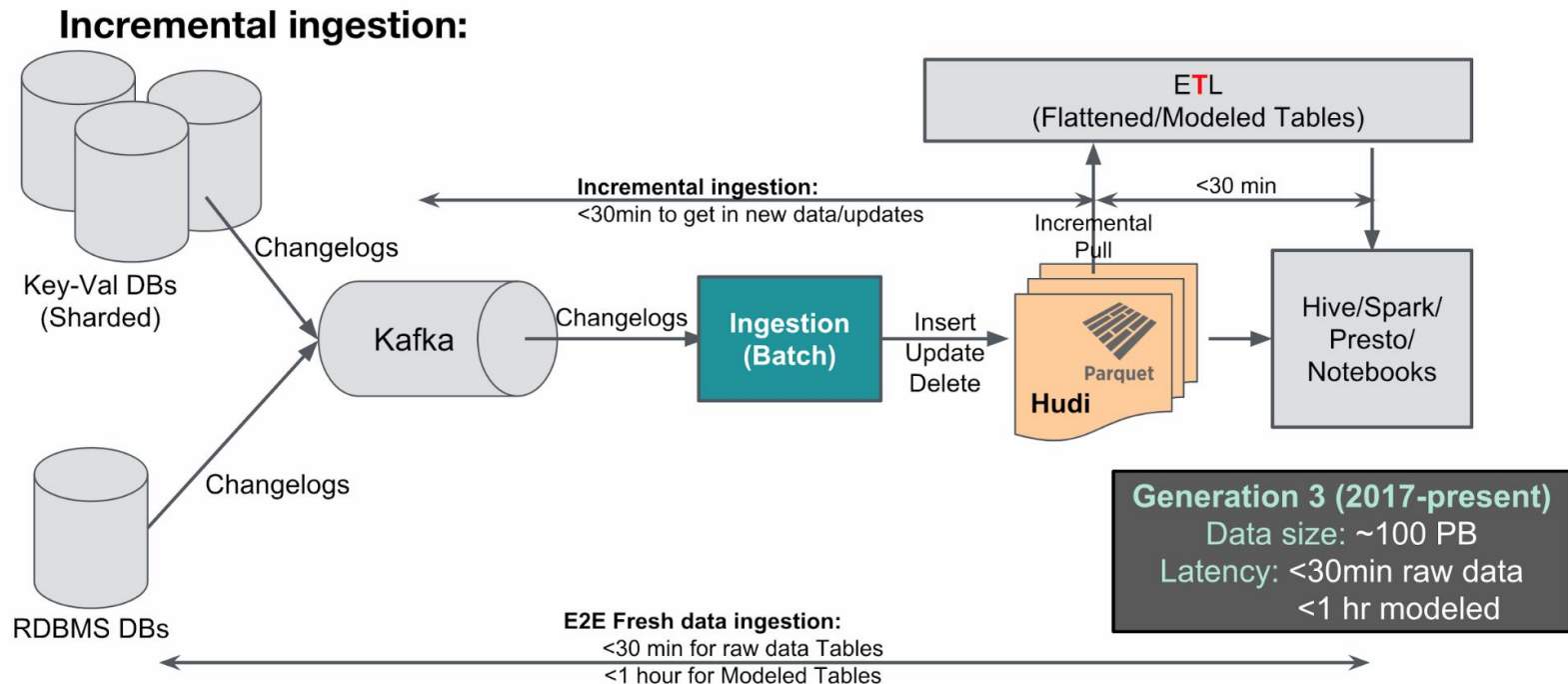
Twitter's data platform (circa 2012)

AIRBNB DATA INFRA



AirBnB's data platform (circa 2016)

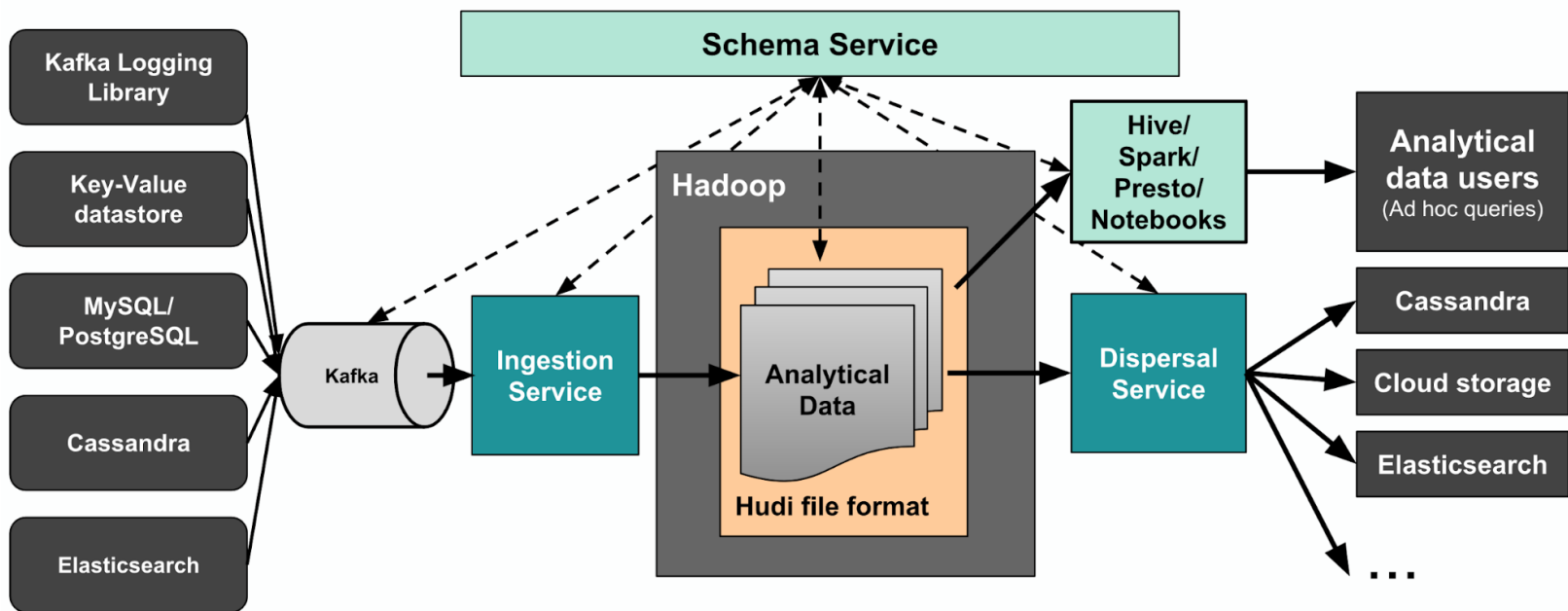
Generation 3 (2017-present) - Let's rebuild for long term



Uber's data platform (circa 2018)

<https://www.uber.com/en-CA/blog/uber-big-data-platform/>

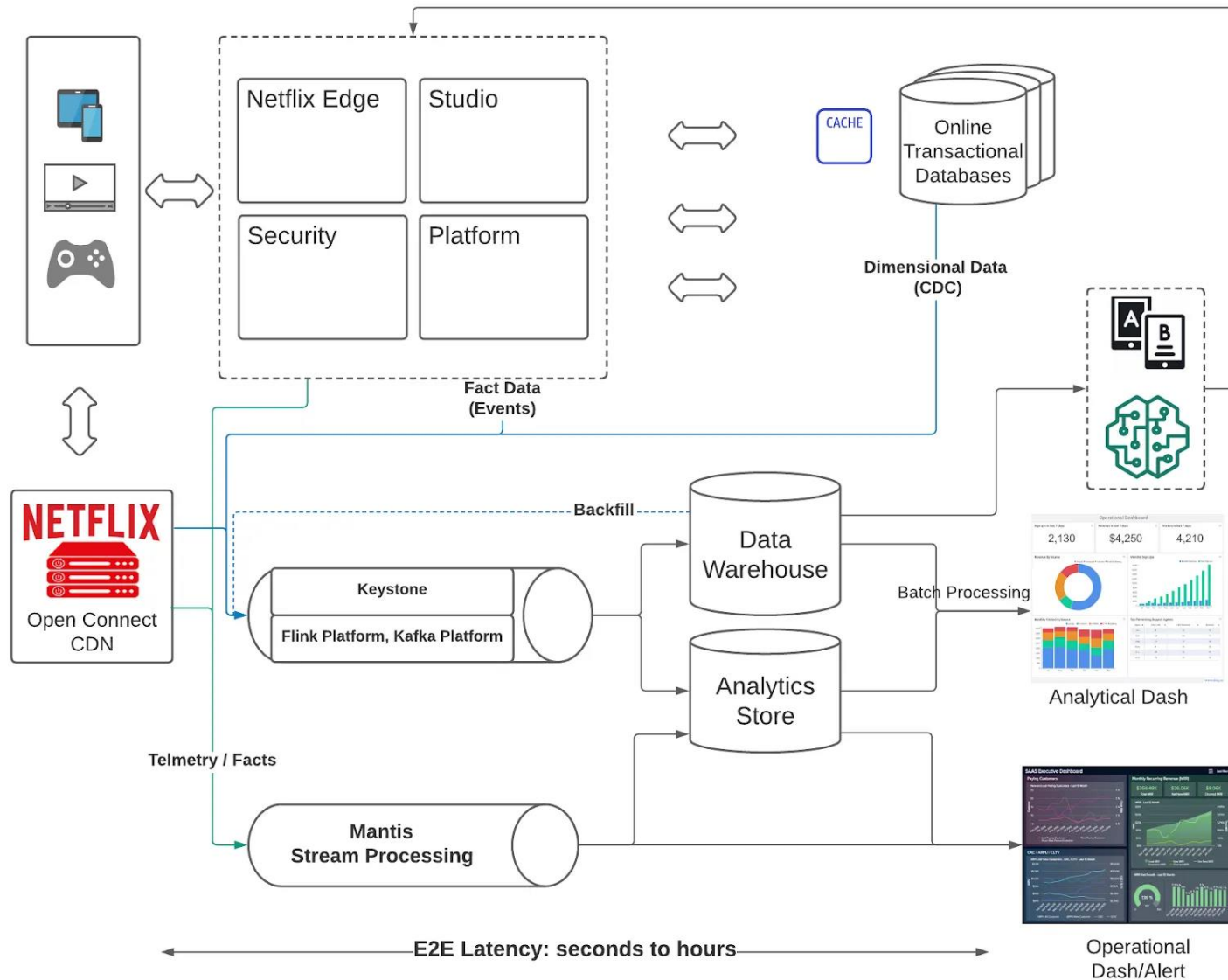
Generic Any-to-Any Data platform



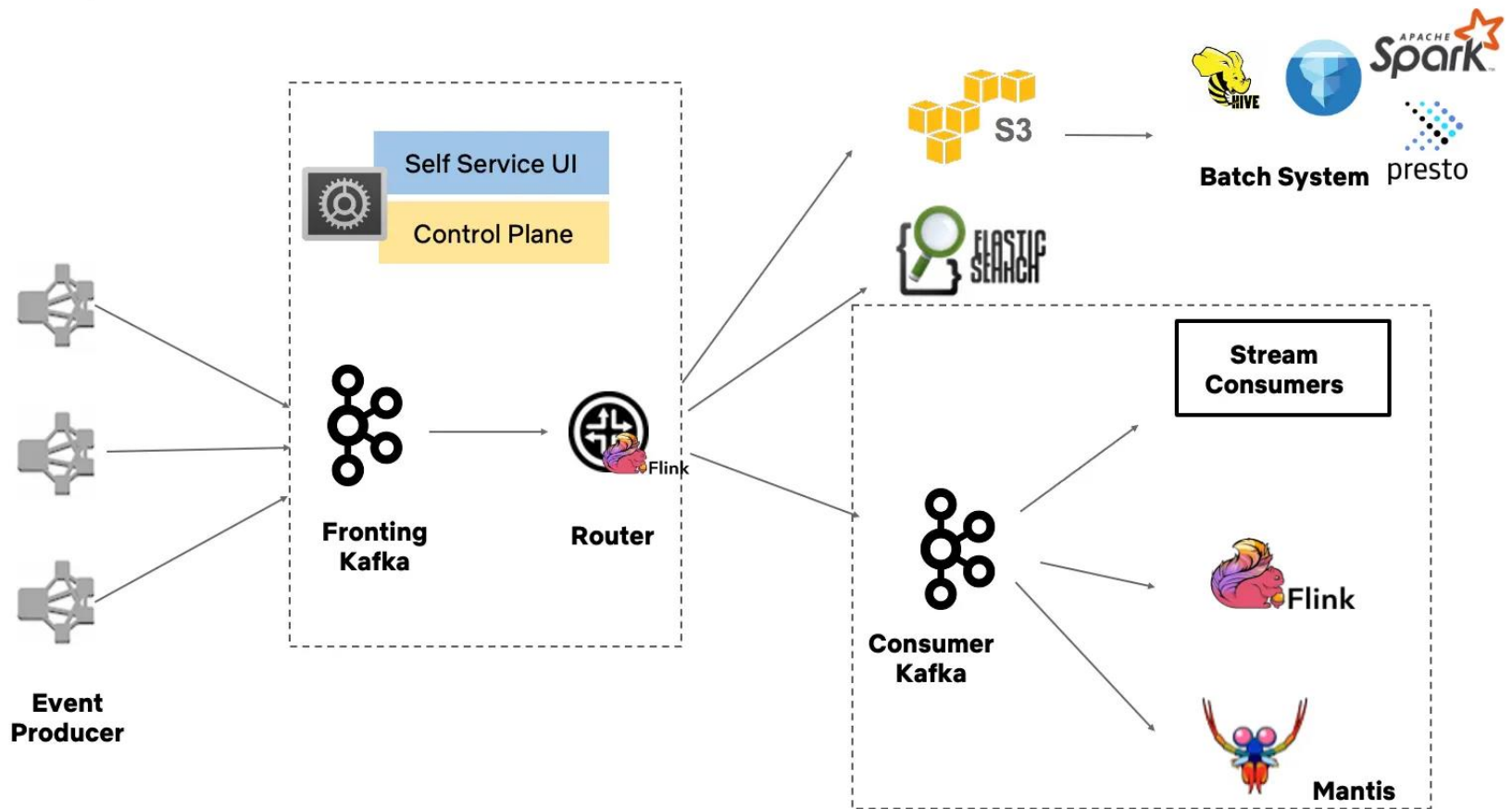
Uber's data platform (circa 2018)

<https://www.uber.com/en-CA/blog/uber-big-data-platform/>

How Stream Processing fit in Netflix (2021)



Netflix's data platform (circa 2021)



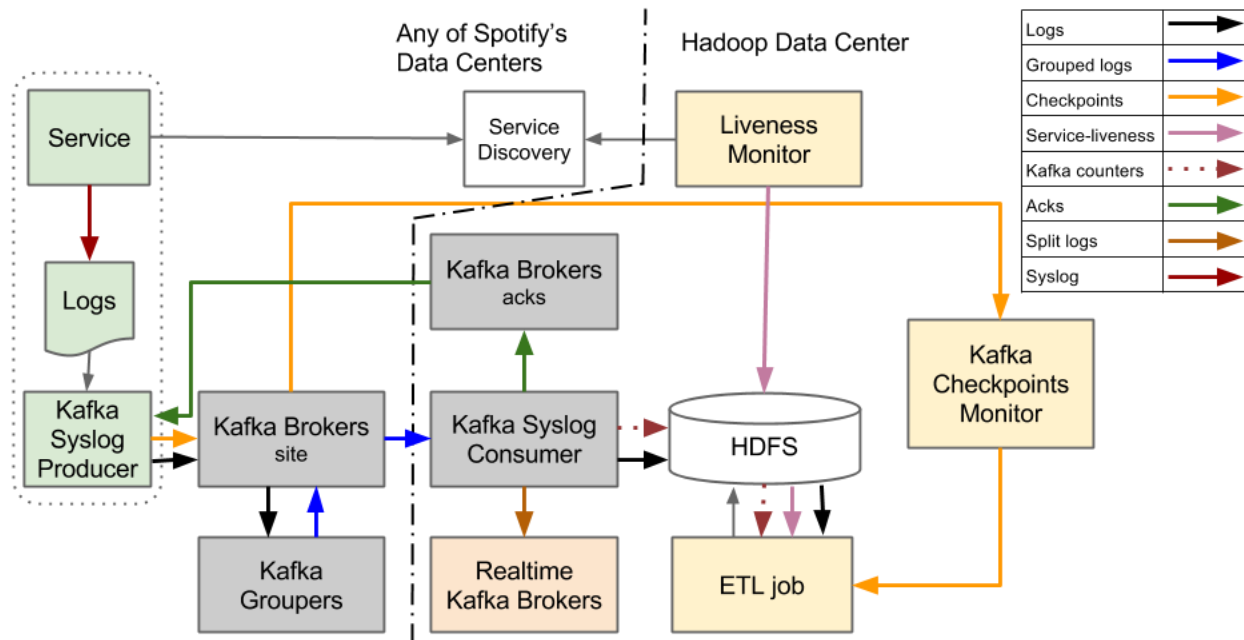
Publish, Collect, Move & Compute event data

Netflix's data platform (circa 2021)



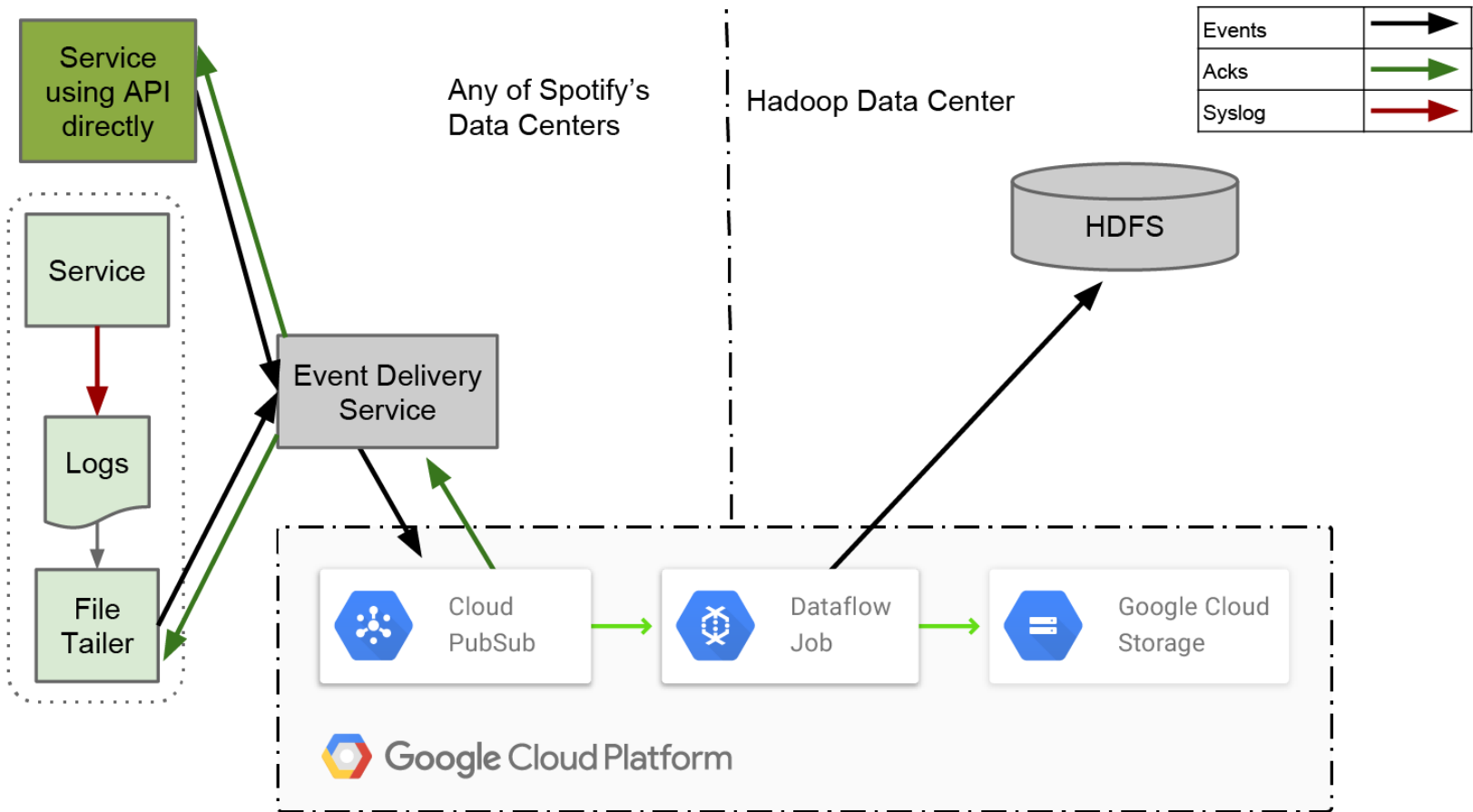
Spotify's data platform (circa 2024)

<https://engineering.atspotify.com/2024/5/data-platform-explained-part-ii>



Spotify's data delivery architecture (circa 2016, pre-cloud)

<https://engineering.atspotify.com/2016/02/spotify-event-delivery-the-road-to-the-cloud-part-i>



Spotify's data delivery architecture (circa 2016, post-cloud)

<https://engineering.atspotify.com/2016/3/spotify-s-event-delivery-the-road-to-the-cloud-part-ii>

富嶽三十六景 神奈川
浪裏

